

Assignment 1 - Easy

Table of contents

1	Instructions	2
1.1	Question 1 (Conceptual)	2
1.2	Question 2 (Conceptual)	2
1.3	Question 3 (Conceptual)	2
1.4	Question 4 (Conceptual)	3
1.5	Question 5 (Conceptual)	3
1.6	Question 6 (Conceptual)	3
1.7	Question 7 (Conceptual)	3
1.8	Question 8 (Conceptual)	4
1.9	Question 9 (Conceptual)	4
1.10	Question 10 (Conceptual)	4
1.11	Question 11 (Calculation)	4
1.12	Question 12 (Calculation)	4
1.13	Question 13 (Calculation)	5
1.14	Question 14 (Calculation)	5
1.15	Question 15 (Calculation)	5
1.16	Question 16 (Calculation)	5
1.17	Question 17 (Calculation)	5
1.18	Question 18 (Calculation)	6
1.19	Question 19 (Calculation)	6
1.20	Question 20 (Calculation)	6
1.21	Question 21 (Spreadsheet)	6
1.22	Question 22 (Spreadsheet)	6
1.23	Question 23 (Spreadsheet)	7
1.24	Question 24 (Spreadsheet)	7
1.25	Question 25 (Spreadsheet)	7

1 Instructions

This assignment is based on Lectures 1-6: bond features and risks, pricing and yield measures, duration and convexity, benchmark spreads and term structure, Treasury and corporate debt markets, and residential mortgages/MBS.

The questions are original CFA-style practice questions written from the course material. They are not copied from CFA Institute books or exams.

Questions 1-10 are conceptual, Questions 11-20 are calculations, and Questions 21-25 use the generated spreadsheet files in this folder.

1.1 Question 1 (Conceptual)

Why does a plain vanilla bond have a predictable cash-flow pattern?

Answer. A plain vanilla bond fixes the coupon rate, coupon dates, face value, and maturity date in the indenture. If the issuer does not default and there is no early redemption, the investor knows both the size and timing of all coupon payments and the final principal repayment. That is why fixed income valuation begins by discounting promised cash flows.

1.2 Question 2 (Conceptual)

Explain the difference between a fixed-rate bond and a floating-rate note.

Answer. A fixed-rate bond pays the same coupon rate through its life, so its price changes when market yields move away from the coupon. A floating-rate note resets its coupon using a reference rate such as SOFR plus a quoted margin, so its income adjusts with short-term rates. Right after reset, a simple floater often trades close to par if credit spreads are unchanged.

1.3 Question 3 (Conceptual)

Why does a callable bond usually offer a higher yield than a similar noncallable bond?

Answer. The call option belongs to the issuer and is valuable when rates fall. The issuer can redeem the old high-coupon bond and refinance at lower rates, which limits the investor's upside and creates reinvestment risk. Investors therefore require extra yield as compensation for giving the issuer that option.

1.4 Question 4 (Conceptual)

What is credit spread risk?

Answer. Credit spread risk is the risk that the yield spread over a benchmark Treasury or spot curve widens. Even if Treasury rates do not move, a wider spread increases the required yield on the corporate bond and lowers its price. This risk is separate from actual default; a bond can lose value simply because the market demands more compensation for credit or liquidity.

1.5 Question 5 (Conceptual)

Why should duration not be interpreted as only a measure of time?

Answer. Macaulay duration has time units, but in risk management duration is mainly a measure of price sensitivity to yield changes. Modified duration translates duration into an approximate percentage price change for a small yield change. Two bonds with similar maturities can have very different durations because coupons, yields, and embedded options change the timing and sensitivity of cash flows.

1.6 Question 6 (Conceptual)

What does positive convexity mean for an option-free bond?

Answer. Positive convexity means the price-yield curve bows outward. For equal-sized yield changes, the price gain when yields fall is larger than the price loss when yields rise. Duration gives the linear first-order effect, while convexity captures the curvature that improves the approximation for larger rate moves.

1.7 Question 7 (Conceptual)

Why is a spot rate curve better than one yield-to-maturity for valuing a coupon bond?

Answer. A coupon bond has cash flows at many dates. Each cash flow should be discounted at the rate appropriate for its own maturity, so spot rates are conceptually correct. A single yield-to-maturity is a summary discount rate that forces all cash flows to use one rate and can hide errors when the curve is steep or humped.

1.8 Question 8 (Conceptual)

What is the taxable-equivalent yield of a municipal bond used for?

Answer. It converts a tax-exempt yield into the pretax yield a taxable bond would need to offer to leave the investor equally well off. The formula is tax-exempt yield divided by one minus the marginal tax rate. It matters because investors compare after-tax returns, not only quoted yields.

1.9 Question 9 (Conceptual)

Why are Treasury bills sold at a discount?

Answer. Treasury bills are zero-coupon securities with maturities of one year or less. They do not pay periodic interest; instead, investors buy them below face value and receive face value at maturity. The difference between purchase price and maturity value is the investor's interest income.

1.10 Question 10 (Conceptual)

What is the main risk unique to mortgage-backed securities relative to simple Treasury notes?

Answer. Mortgage-backed securities contain borrower prepayment options. When rates fall, borrowers refinance and return principal early, forcing investors to reinvest at lower yields. When rates rise, prepayments slow and the security extends, leaving investors exposed to a longer-duration asset than expected.

1.11 Question 11 (Calculation)

A 3-year annual-pay bond has face value \$1,000, a 5% coupon, and a 6% yield. Compute the price.

Answer. The annual coupon is \$50. $\text{Price} = \$50/1.06 + \$50/1.06^2 + \$1,050/1.06^3 = \973.27 . Because the coupon rate is below the yield, the bond sells below par.

1.12 Question 12 (Calculation)

Compute the price of a 3-year zero-coupon bond with \$1,000 face value and a 4% annual yield.

Answer. The only cash flow is the \$1,000 maturity payment. $\text{Price} = \$1,000/(1.04)^3 = \889.00 .

1.13 Question 13 (Calculation)

A quarterly floater pays SOFR + 150 bps. SOFR at reset is 3.50%. Face value is \$1,000. What is the next coupon payment?

Answer. The annual coupon rate is $3.50\% + 1.50\% = 5.00\%$. Quarterly payment = $5.00\%/4 \times \$1,000 = \12.50 .

1.14 Question 14 (Calculation)

A semiannual coupon bond pays \$25 every six months. Settlement occurs 30 days after the last coupon in a 180-day coupon period. Compute accrued interest.

Answer. Accrued interest = $\$25 \times 30/180 = \4.17 . The buyer pays this amount to compensate the seller for interest earned since the last coupon date.

1.15 Question 15 (Calculation)

A 5-year corporate bond yields 5.20% and the 5-year Treasury yields 3.90%. Compute the benchmark spread in basis points.

Answer. Spread = $5.20\% - 3.90\% = 1.30\%$, or 130 basis points.

1.16 Question 16 (Calculation)

A tax-exempt municipal bond yields 3.20%. The investor's marginal tax rate is 30%. Compute the taxable-equivalent yield.

Answer. Taxable-equivalent yield = $3.20\% / (1 - 0.30) = 4.57\%$. A taxable bond must yield about 4.57% to match the municipal bond after tax.

1.17 Question 17 (Calculation)

A 91-day Treasury bill with \$10,000 face value is quoted at a 4.75% bank discount yield. Compute the price.

Answer. Price = $\$10,000 \times [1 - 0.0475 \times 91/360] = \$9,879.93$.

1.18 Question 18 (Calculation)

A Treasury note is quoted at 98-24. For \$1,000 face value, compute the clean price.

Answer. 98-24 means $98 + 24/32 = 98.75\%$ of par. Clean price = $98.7500\% \times \$1,000 = \987.50 .

1.19 Question 19 (Calculation)

An amortizing security repays principal of \$100, \$150, \$250, and \$500 at years 1-4. Compute weighted average life.

Answer. $WAL = (100 \times 1 + 150 \times 2 + 250 \times 3 + 500 \times 4) / 1,000 = 3.15$ years.

1.20 Question 20 (Calculation)

Convert a 6% CPR into a single monthly mortality rate.

Answer. $SMM = 1 - (1 - 0.06)^{(1/12)} = 0.5143\%$. This is the monthly prepayment rate applied after scheduled principal.

1.21 Question 21 (Spreadsheet)

Open `easy_bond_prices.csv`. Use the spreadsheet to compute each bond's annual coupon dollar amount and decide whether it trades at a discount, premium, or par based only on coupon rate versus yield.

Answer. The coupon dollars are face value multiplied by coupon rate. Bond A: \$40 and discount because $4.00\% < 5.00\%$. Bond B: \$60 and premium because $6.00\% > 4.50\%$. Bond C: \$50 and par because $5.00\% = 5.00\%$.

1.22 Question 22 (Spreadsheet)

Open `easy_spreads.csv`. Use the spreadsheet to compute the spread over Treasury for each issuer in basis points.

Answer. Subtract the Treasury yield from the issuer yield and multiply by 10,000. Agency = 35 bps, Utility = 95 bps, High Yield Industrial = 325 bps. The ranking shows how credit and liquidity risk increase required yield.

1.23 Question 23 (Spreadsheet)

Open `easy_accrued_interest.csv`. Use the spreadsheet to compute accrued interest for each bond.

Answer. Accrued interest equals coupon payment times days since last coupon divided by days in coupon period. Bond A = \$6.25, Bond B = \$11.67, Bond C = \$16.67.

1.24 Question 24 (Spreadsheet)

Open `easy_mortgage_prepayments.csv`. Use the spreadsheet to compute SMM and the expected prepayment for each row.

Answer. For each CPR, $SMM = 1 - (1 - CPR)^{(1/12)}$, then prepayment = SMM x (beginning balance - scheduled principal). The answers are approximately \$2,568, \$4,347, and \$7,129 for months 1-3.

1.25 Question 25 (Spreadsheet)

Open `easy_taxable_equivalent.csv`. Use the spreadsheet to compute taxable-equivalent yield for each municipal bond.

Answer. Taxable-equivalent yield equals municipal yield divided by one minus the tax rate. The yields are 4.29%, 4.62%, and 5.00%, respectively.